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1NRI

APPLIED (ENTREPRENEURSHIP) PROJECT

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Chapter 1: Introduction

1.1 Problem Overview

Globally, the E-commerce landscape has experienced exponential growth. Tohidi et al [1] provide evidence of a 'notable shift' in consumer preferences towards online sales, hence the expansion of the e-commerce market. Despite this, small to medium-sized enterprises (SMEs) in emerging markets, such as Africa, have faced several significant barriers to their digital transformation. These challenges are well-documented, as per the Trade and Development wing of the United Nations [2]. This disparity is even more apparent in the fashion sector, where 17% of SMEs achieve full digital tool adoption, compared to 54% of larger brands [3]. This technological divide creates competitive disadvantages for these SMEs when coming up against larger, more established fashion brands within sub-Saharan Africa.

Fashion SMEs work in a data and technology-intensive environment. Their business operations span from supply chain management for fabric sourcing to social media marketing to promote brand visibility, inventory control for managing stock, and e-commerce platforms as a primary, accessible sales channel. Even though these businesses rely on digital tools for all these operations, the valuable data gathered from these channels tends to be fragmented. As a result, combined insights cannot be obtained from the data to push data-driven decision-making. Larger, more established brands are able to utilise Enterprise Resource Planning (ERP) systems. These ERP systems are integrated software solutions designed to manage and automate core business functions within an organisation, encompassing finance, supply chain, manufacturing, human resources, and customer relationship management [4].

For larger brands, comprehensive ERP solutions with seamless E-commerce integration have enabled holistic operational visibility [5]. However, for SMEs, adoption of such tools is not as common. Among the barriers to adoption, prohibitive costs, a lack of localised solutions, inadequate technical support infrastructure, and limited integration with regional payment gateways are a few that stand out among key stakeholders.

This paper presents the development of IRIS, an integrated e-commerce and ERP platform specifically designed for 1NRI Worldwide Ltd., a Ghanaian fashion SME. Our custom solution addresses data fragmentation by consolidating information from multiple operational channels, including social commerce platforms, inventory systems, and marketing tools, into a unified analytics dashboard that provides real-time business intelligence.

1.2 Problem Background

SMEs in the fashion sector face distinct operational challenges that differentiate them from traditional retail businesses. The fashion supply chain involves multiple stages: fabric and material sourcing, production coordination, quality control, inventory management, marketing through social media channels, and customer acquisition via e-commerce platforms. Each stage generates valuable data, yet this information remains siloed across disconnected systems [6].

Many of these businesses in the African context rely on ad-hoc digital tools, such as WhatsApp Business for customer communication, Instagram and TikTok for brand marketing, third-party payment processors like Flutterwave and Paystack, and tools like spreadsheets or more established systems like QuickBooks for inventory management.

Whilst these tools bring great value as standalone systems, owners of Fashion SMEs are unable to derive the extent to which decisions made drive sales, trends within the depletion rates across inventory items, and other occurrences which inform strategic decision making.

Data fragmentation represents the central challenge facing the average fashion SME seeking digital transformation. The success of E-commerce-based brands relies heavily on data-driven insights from the aforementioned sections of their operation [7]. However, the tools at their disposal are often used disjointly, eliminating the possibility for a centralised pool of data from which business intelligence can be adequately generated and utilised. On the other hand, the ERP systems available are not targeted toward SMEs, but rather for larger businesses with dedicated, sizeable budgets.

1.3 Related Work

1.3.1 E-commerce Analytics Platforms

The evolution of e-commerce analytics platforms has been pivotal in shaping the digital transformation of retail businesses globally. State-of-the-art solutions, such as Triple Whale, exemplify the integration of multi-channel data consolidation, advanced attribution modeling, and predictive analytics for inventory and customer management, serving over 50,000 brands worldwide [8]. These platforms enable businesses to optimise operations by aggregating data from diverse sources, ranging from sales channels and advertising platforms to inventory systems, into unified dashboards that facilitate actionable insights and strategic decision-making [8–10].

However, the adoption of such platforms in African markets faces significant challenges. High subscription costs, often exceeding the revenue thresholds of most African fashion SMEs, and a lack of integration with local payment systems and mobile money platforms (such as MTN MoMo, Flutterwave Store, Paystack Commerce, and Hubtel) limit their accessibility and utility

[11–13]. African-focused digital platforms have emerged to address these gaps, offering tailored solutions that integrate with regional payment ecosystems and support local business models [11, 13]. For example, platforms like Oze provide SME analytics specifically designed for African entrepreneurs, while Jumia and other regional players have developed digital infrastructures that accommodate the unique needs of African e-commerce, including mobile money integration and localised logistics [12, 13, 11].

Recent research emphasises the significance of context-specific analytics and the necessity for platforms that cater to the operational realities of African SMEs, including lower average transaction values, informal business practices, and the prevalence of mobile-first commerce [11, 13, 14]. The proliferation of digital platforms in Africa is further supported by the rapid expansion of mobile money and fintech ecosystems, which have become critical enablers of e-commerce adoption and financial inclusion across the continent [11, 13, 15, 16].

1.3.2 Digital Transformation in Fashion

Digital transformation in the fashion industry is characterised by the adoption of technologies that enhance supply chain transparency, enable real-time inventory management, and support data-driven production planning [17–19]. Studies have shown that digital platforms facilitate operational agility, allowing fashion businesses to respond dynamically to shifting market demands and consumer preferences [17, 18, 20]. Key success factors include the integration of supply chain data, personalised marketing through customer data platforms, and the implementation of sustainability tracking systems [17–20].

While much of the literature focuses on established fashion enterprises in developed markets, there is a growing body of research examining the digitalisation of African fashion supply chains. These studies underscore the importance of platforms that accommodate the unique characteristics of the fashion sector - such as seasonal product cycles, style and trend sensitivity, and the complexity of size and colour variants [17–20].

African fashion SMEs, in particular, benefit from digital solutions that are tailored to their operational contexts, including mobile-based inventory management, localised e-commerce platforms, and integration with regional logistics providers [11, 13, 21].

Empirical evidence from Nigeria and other African markets demonstrates that e-commerce adoption significantly enhances the profitability and competitiveness of fashion SMEs, provided that digital infrastructure and logistics are strategically integrated [21]. The emergence of digital platforms like Selar and the expansion of mobile money-driven commerce in countries such as Ghana, Kenya, and Nigeria further illustrate the transformative potential of digital technologies in the African fashion industry [11, 13, 15].

1.3.3 ERP Solutions for African Markets

The implementation of global ERP solutions in African markets has been extensively analyzed, revealing persistent barriers related to technical support, integration with local financial systems,

and regulatory compliance [22, 11, 23]. Platforms such as Microsoft Dynamics and Odoo, despite their comprehensive feature sets, often struggle to achieve widespread adoption among African SMEs due to misalignments with local business practices and infrastructural constraints [22, 11, 23].

A critical challenge is the limited availability of localised technical support, with most ERP vendors maintaining a minimal physical presence on the continent, thereby necessitating reliance on remote assistance that may lack contextual understanding [11, 23]. Integration with African mobile money systems and regional banks remains limited, frequently requiring costly custom development and undermining the core value proposition of unified data management [11, 23]. Additionally, multi-currency functionality and compliance features are often designed for Western regulatory environments, necessitating significant customisation to address the complexities of cross-border African trade and informal exchange rates [11, 23].

Recent research advocates for the development of ERP solutions specifically architected for African SMEs, emphasizing the need for platforms that integrate seamlessly with local payment ecosystems, support multi-currency transactions, and provide robust reporting tailored to regional regulatory requirements [11, 23, 13]. The rise of African-focused digital platforms and the increasing adoption of mobile money-driven ERP functionalities signal a shift towards more contextually relevant solutions that address the unique operational realities of African businesses [11, 13, 15, 16].

Collectively, these findings highlight the necessity of specialised digital platforms that combine e-commerce and ERP capabilities, designed to meet the specific needs of fashion SMEs in Sub-Saharan Africa. Such platforms must address the shortcomings of global enterprise solutions by offering localised integration, affordable pricing models, and support for the diverse payment and regulatory environments prevalent across the continent [11, 13, 23, 21].

Chapter 2: Requirements Analysis

2.1 Introduction

The requirement analysis for IRIS addresses the critical operational challenges facing fashion SMEs in Sub-Saharan Africa, particularly data fragmentation, inadequate business intelligence, and limited access to affordable, localised digital solutions. This chapter delineates both functional and non-functional requirements derived from stakeholder analysis, market research, and the specific operational needs of 1NRI Worldwide Ltd, as a representative fashion SME.

Requirements analysis forms the foundation of successful software development, particularly for enterprise systems like ERP platforms [23, 24]. Functional requirements define what a system must do and what its features and functions are, while non-functional requirements describe the general properties and quality attributes of a system [27]. For ERP systems specifically, gathering comprehensive requirements is crucial, as studies by Gartner indicate that the top reason for ERP implementation failure is a lack of understanding of these requirements [29].

The requirements framework is structured to ensure the platform delivers a unified solution that consolidates disparate data sources, provides actionable insights, and integrates seamlessly with the regional digital ecosystem. Each requirement directly addresses one or more of the key challenges identified in Chapter 1.

2.2 Stakeholder Analysis

The primary stakeholders for the 1RIS include:

- Business Owners and Management: Require comprehensive visibility into business operations, real-time analytics for strategic decision making, and consolidated reporting across all business functions. Their needs centre on understanding sales trends, inventory performance, marketing ROI, and overall business health through intuitive dashboards.
- Operations Staff: Need efficient tools for inventory management, order processing, and supply chain coordination. They require streamlined workflows that reduce manual data entry and facilitate seamless communication between procurement, production, and sales functions.
- Marketing Personnel: Require integration with social media platforms (Instagram, TikTok, Facebook), tracking of marketing campaign performance, customer engagement metrics, and the ability to analyse which marketing efforts drive actual sales conversions.
- Customers: Expect a reliable, user-friendly e-commerce experience with multiple local payment options, order tracking capabilities, and responsive customer service. Their requirements include mobile-optimised interfaces and support for popular African

payment methods
- Financial stakeholders: Need accurate financial reporting, multi-currency support, integration with local accounting standards, and automated reconciliation between sales channels and financial records.

2.3 Functional Requirements

Functional requirements define the specific capabilities and features IRIS must provide to address the operational needs of fashion SMEs. Each requirement is mapped to the core problems of data fragmentation and limited digital integration identified in the problem background.

FR1: Multi-Channel Sales Integration

- The platform must aggregate sales data from multiple channels, including the native e-commerce storefront, social commerce platforms (such as Instagram Shop, Facebook Marketplace, and WhatsApp Business), and physical retail locations, into a unified sales database.

FR2: Integrated Inventory Management System

- The platform must provide comprehensive inventory tracking with support for fashion-specific attributes, including size, colour, style variants, fabric types, and seasonal collections. The system must automatically update inventory levels across all sales channels when stock changes occur.

FR3: Local Payment Gateway Integration

- The platform must integrate with major African payment processors, including MTN Mobile Money, Flutterwave, Paystack, and Hubtel, and support multiple currencies (GHS, NGN, KES, USD, CAD) with real-time exchange rate updates.

FR4: Unified Analytics Dashboard

- The platform must provide a comprehensive analytics dashboard that consolidates metrics from sales, inventory, marketing, and customer data, along with customizable reports and data visualisation capabilities.

FR5: Supply Chain and Procurement Management

- The platform must facilitate supplier management, purchase order creation, goods receipt tracking, and supplier performance analytics to streamline the procurement process from fabric sourcing to finished goods

FR6: Customer Relationship Management (CRM)

- The platform must maintain comprehensive customer profiles aggregating data from all touchpoints, including purchases, social media interactions, customer service inquiries, and marketing engagement

FR7: Role-Based Access Control and Multi-User Support

- The platform must support multiple user accounts with granular permissions control

2.4 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints that ensure the platform is viable, reliable, and sustainable for fashion SMEs operating in African markets.

NFR1: Performance and Scalability

- The platform must maintain responsive performance under typical SME workloads and scale gracefully as the business grows from startup to established enterprise.

NFR2: Reliability and Availability

- The platform must maintain high availability with minimal unplanned downtime, automated backup systems, and disaster recovery capabilities to protect critical business data.

NFR3: Mobile Responsiveness and Cross-Platform Compatibility

- The platform must provide fully functional interfaces optimised for mobile devices, recognising that mobile commerce dominates African e-commerce and many SME owners manage their businesses primarily via smartphones.

NFR4: Security and Data Protection

- The platform must implement comprehensive security measures that protect customer payment information, business financial data, and proprietary information, while also complying with relevant data protection regulations.

NFR5: Usability and User Experience

- The platform must prioritise intuitive user interfaces requiring minimal training, with contextual help, guided workflows, and progressive disclosure of complexity to accommodate users with varying technical proficiency.

NFR6: Cost Effectiveness and Pricing Model

- The platform must offer a pricing structure that is affordable to fashion SMEs with monthly revenues between $1,000 and $50,000, avoiding the prohibitive costs of enterprise ERP solutions while maintaining financial sustainability.

NFR7: Integration and Extensibility

- The platform must provide well-documented APIs and webhook systems enabling integration with third-party services, future feature additions, and customisation to accommodate evolving business needs without core system modifications.

2.5 Requirements Traceability

The functional and non-functional requirements specified in this chapter directly address the core challenges identified in Chapter 1:

Data Fragmentation: FR1 (Multi-Channel Sales Integration), FR4 (Unified Analytics Dashboard), and FR6 (CRM) consolidate scattered data sources into unified repositories, eliminating the silos that prevent holistic business insight.

Prohibitive Costs: NFR6 (Cost Effectiveness) ensures the solution is financially accessible to SMEs, with pricing 60-70% lower than traditional ERP systems while maintaining comprehensive functionality.

Local Payment Integration: FR3 (Local Payment Gateway Integration) addresses the critical gap in global platforms by supporting mobile money and regional payment processors, which are essential for African e-commerce.

Fashion-Specific Operations: FR2 (Integrated Inventory Management) and FR5 (Supply Chain Management) accommodate the unique complexities of fashion retail, including variant management, seasonal collections, and fabric sourcing workflows.

Technical Accessibility: NFR3 (Mobile Responsiveness) and NFR5 (Usability) ensure the platform is accessible to users with limited technical expertise and those primarily operating via mobile devices, reflecting African market realities.

The subsequent chapters will detail the system architecture, implementation approach, and validation methodologies, demonstrating how these requirements are fulfilled in the delivered solution for 1NRI Worldwide Ltd.

2.6 Requirement Prioritisation

Using the MoSCoW method (Must Have, Should Have, Could Have, Won't have), requirements are prioritised for the minimum viable product (MVP) and subsequent iterations. The MoSCoW method is a widely adopted prioritisation technique in agile software development, originally developed by Dai Clegg at Oracle to help teams prioritise tasks during product releases [26, 29].

Must Have (MVP - Phase 1):

- FR1: Multi-Channel Sales Integration (core to solving data fragmentation)
- FR2: Integrated Inventory Management (fundamental operational need)
- FR3: Local Payment Gateway Integration (critical for customer transactions)
- FR7: Role-Based Access Control (essential for security and growth)
- NFR2: Reliability and Availability
- NFR4: Security and Data Protection (non-negotiable for handling financial data)

Should Have (Phase 2):

- FR4: Unified Analytics Dashboard
- FR6: Customer Relationship Management
- NFR1: Performance and Scalability
- NFR5: Usability and User Experience

Could Have (Phase 3):

- FR5: Supply Chain and Procurement Management
- NFR3: Advanced Mobile Features
- NFR7: Extensive Integration and Extensibility

This phased approach ensures rapid delivery of core value while establishing a foundation for systematic feature expansion aligned with user feedback and business growth.

Chapter 3: System Architecture and Design

3.1 Introduction

This chapter presents the comprehensive system architecture and design for IRIS. The architecture is designed to address the specific requirements identified in Chapter 2, with particular emphasis on solving data fragmentation, ensuring cost-effectiveness, and providing seamless integration with African payment ecosystems and social commerce platforms. The architecture also prioritizes modularity, scalability, and maintainability to support the platform's evolution from an MVP (Minimum Viable Product) to a comprehensive enterprise solution.

3.2 Architectural Principles and Patterns

The platform's architecture is guided by the following core principles:

1. Separation of Concerns: The system is structured into distinct layers (presentation, business logic, data access) to enable independent development, testing, and scaling of components
2. Microservices-Oriented Design: The architecture employs service-oriented principles to enable independent deployment and scaling of critical functions such as payment processing, inventory management, and analytics
3. API-First Design: All system functionality is exposed through well-defined RESTful APIs, enabling integration with third-party services, future mobile applications, and partner systems
4. Data centralization: A unified database eliminates data silos across sales channels, inventory, and customer interactions

3.3 High-Level System Architecture

IRIS employs a three-tier architecture with microservice characteristics for critical business domains.

3.3.1 Presentation Layer

The presentation layer consists of:

- **Customer-Facing E-commerce Interface:** A responsive web application optimized for mobile devices, supporting product browsing, shopping cart, checkout with local payment methods, and order tracking
- **Business Management Dashboard:** An interface for inventory management, order processing, analytics visualization, customer relationship management, and supplier coordination.

Both interfaces are built using modern web technologies with server-side rendering for optimal performance and SEO.

3.3.2 Application Layer

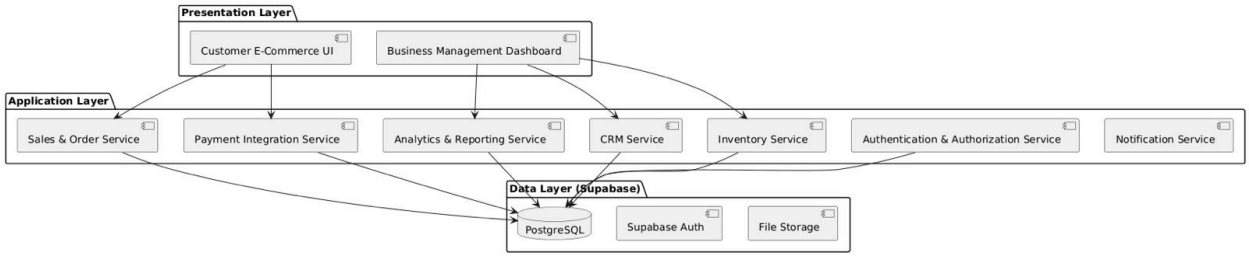
The application layer implements core business logic through modular service components:

1. **Sales and Order Management Service:** Handles multi-channel order processing, order lifecycle management, and sales data consolidation from e-commerce and social commerce platforms (FR1)
2. **Inventory Management Service:** Manages product catalog with fashion-specific attributes, real-time stock tracking across channels, automated reorder alerts, and inventory valuation (FR2)
3. **Payment Integration Service:** Provides a unified interface to multiple African payment gateways, including MTN Mobile Money and Paystack. It also handles payment processing, reconciliation, and multi-currency support (FR3)
4. **Analytics and Reporting Service:** Aggregates data from all system modules to generate real-time business intelligence
5. Customer Relationship Management Service: Maintains comprehensive customer profiles, tracks interactions across touchpoints, segments customers for targeted marketing (FR6)
6. **Supply Chain Management Service:** Coordinates supplier relationships, purchase orders, goods receipt, and supplier performance tracking. This is scheduled for Phase 3 implementation (FR5)
7. **Authentication and Authorization Service:** Centralized identity management implementing OAuth, role-based access control, and session management (FR7)
8. **Notification Service:** Handles email, SMS, and push notifications for order confirmations, shipping updates, and low stock alerts.

3.3.3 Data Layer

The data layer utilizes Supabase, an open-source Backend-as-a-Service (BaaS) built on PostgreSQL, providing a PostgreSQL database, real-time subscriptions, authentication, storage, and row-level security

Here is the compiled System Architecture Diagram:



3.4 Database Design

The database schema models the essential domain entities required for integrated e-commerce and ERP operations.

3.4.1 Entity Design Overview

User Management Entities:

- **users:** Stores user accounts with authentication credentials and profile information
- **roles:** Define system roles (admin, manager, staff, customer)
- **user_roles:** Maps users to roles for role-based access control

Product and Inventory Entities:

- **products:** Master product catalog with basic information
- **product_variants:** Manages fashion-specific variants (size, color, style)
- **inventory:** Tracks stock levels, location, and valuation
- **suppliers:** Maintains supplier information and performance metrics

Sales and Order Entities:

- **orders:** Captures customer order with status tracking
- **Order_items:** Link items for each order with product variants
- **payments:** Records payment transactions across multiple gateways
- **sales_channels:** Defines sales (website, Instagram, WhatsApp)

Customer Relationship Entities:

- **customers:** customer profiles with contact information
- **customer_interactions:** Logs all customer touchpoints and communications
- **customer_segments:** Groups customers for targeted marketing

Financial Entities

- **transactions:** records all financial transactions
- **currencies:** Supports multi-currency operations with exchange rates

3.4.2 Schema Design

Authentication Layer: All user-related tables leverage Supabase Auth with:

- UUID primary keys for security and scalability.
- Encrypted password storage using bcrypt.
- Session management with JWT tokens.
- Audit trails with created_at and updated_at timestamps.

Product Catalog Modeling: The product schema captures over 15 attributes including:

- SKU, name, description, category.
- Fashion-specific fields (fabric type, season, collection).
- Pricing with support for multiple currencies.
- Variant management (size, color, style combinations).

Order Tracking Mechanism: Each order implements:

- Foreign keys to customers and sales channels.
- Status workflow (pending, confirmed, processing, shipped, delivered, cancelled).
- Payment reconciliation across multiple gateways.
- Timestamps for order lifecycle events.

3.4.3 Relational Mapping

Core relationships enforced through foreign key constraints:

User-Order Relationships:

- One customer → many orders (1:N)
- One order → many order items (1:N)
- One staff member → many managed orders (1:N)

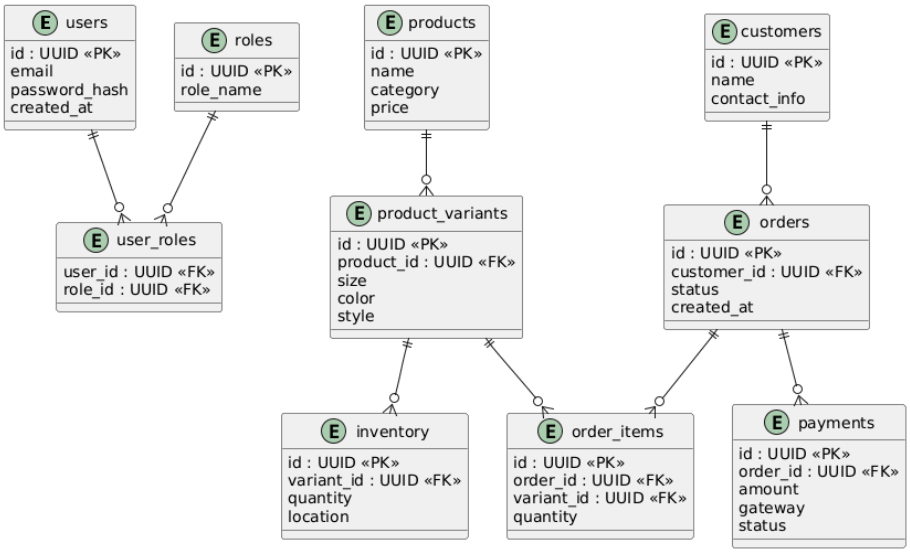
Product-Inventory Relationships:

- One product → many variants (1:N)
- One variant → many inventory records (1:N for multi-location)
- One supplier → many products (1:N)

Payment Relationships:

- One order → many payments (1:N for partial payments)
- One payment gateway → many transactions (1:N)

Here is the full Entity-Relationship diagram:



3.4.4 Security and Compliance Controls

Row-Level Security (RLS):

- Supabase RLS policies ensure customers only access their own orders.
- Staff can only view/edit data within their authorization scope.
- Financial data restricted to authorized personnel.

Data Integrity Constraints:

- NOT NULL constraints on critical fields (email, order status, payment amount).
- UNIQUE constraints on SKU, customer email, transaction IDs.
- CHECK constraints for valid price ranges and inventory quantities.
- CASCADE rules for referential integrity.

Audit and Compliance:

- All tables include a created_at and updated_at field
- Soft deletes with deleted_at for data retention compliance
- Transaction logs for financial operations and inventory changes

3.5 Client-Server Architecture

IRIS adopts a modern client-server architecture with the following characteristics:

Frontend Application:

- Server-side rendered Next.js application for optimal performance and SEO.
- Client-side state management for responsive user experience.
- Progressive Web App (PWA) capabilities for mobile users.

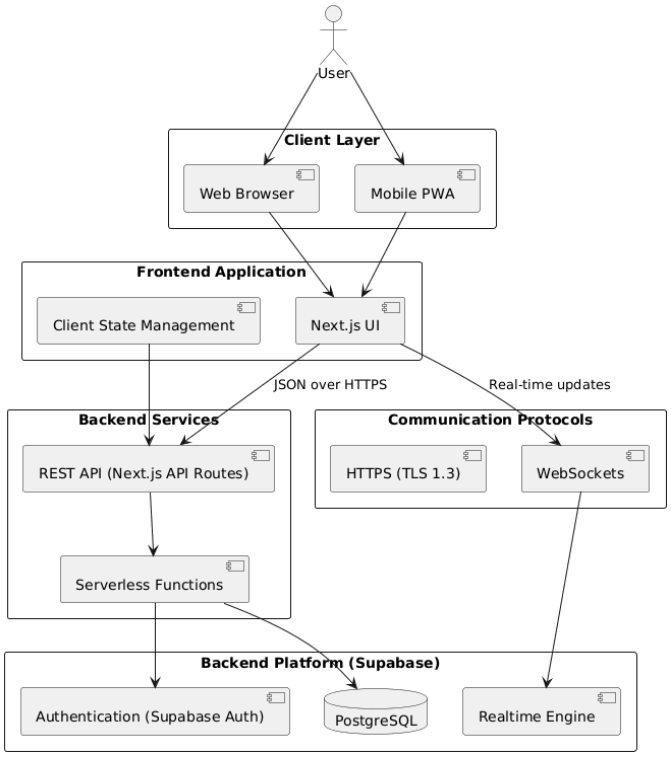
Backend Services:

- RESTful API built with Next.js API routes.
- Serverless functions for compute-intensive operations.
- Supabase backend for database operations and authentication.

Communication Protocol:

- HTTPS for all client-server communication with TLS 1.3 encryption.
- WebSocket connections for real-time inventory and order updates.
- RESTful API endpoints following OpenAPI 3.0 specification.
- JSON data format for API requests and responses.

The full Client-Server Architecture is summarised below:



3.6 Authentication, Authorization, and Security

Authentication Mechanism:

- Email/password authentication via Supabase Auth.
- OAuth integration for social login (Google, Facebook) in future phases.
- JWT-based session tokens with automatic refresh.
- Multi-factor authentication (MFA) for administrative accounts.

Authorization Framework:

- Role-based access control (RBAC) with granular permissions.
- Resource-level permissions for sensitive operations.
- API key authentication for third-party integrations.

Security Measures:

- End-to-end encryption for payment information.
- OWASP Top 10 compliance for web application security.
- Regular security audits and penetration testing.
- DDoS protection and rate limiting on API endpoints.
- Content Security Policy (CSP) headers to prevent XSS attacks.

3.7 Process Architecture

Order Fulfillment Workflow (Appendix 1):

1. Customer places an order through the e-commerce interface or social commerce.
2. The system validates inventory availability and reserves the stock.
3. Payment gateway processes transaction.
4. Order confirmed and notification sent to the customer and staff.
5. Inventory automatically updated across all channels.
6. Order assigned to fulfillment staff.
7. Shipping details are captured, and tracking information is shared.
8. Order marked complete upon delivery confirmation.

Inventory Management Workflow (Appendix 2):

1. Stock levels are monitored in real-time across all sales channels.
2. Automated low-stock alerts trigger reorder notifications.
3. Purchase orders are created and sent to suppliers.
4. Goods receipt recorded upon delivery.

5. Inventory levels updated and reconciled.
6. Variance reports generated for audit purposes.

Analytics Generation Workflow (Appendix 3):

1. Transactional data captured from all system modules.
2. Data is aggregated and transformed in the analytics service.
3. Metrics calculated (sales trends, inventory turnover, customer lifetime value).
4. Dashboards are updated in real-time via WebSocket connections.
5. Scheduled reports generated and delivered to stakeholders.

3.8 Integration Architecture

Payment Gateway Integration:

- Abstraction layer supporting multiple payment processors.
- Webhook handlers for payment confirmations and failures.
- Automatic payment reconciliation with order records.

Social Commerce Integration:

- API integrations with Instagram Shopping and Facebook Marketplace.
- Webhook listeners for social platform orders.
- Automated order import and inventory synchronization.

Third-Party Service Integration:

- Email service providers (SendGrid) for transactional emails.
- SMS gateways for order notifications and marketing.
- Shipping and logistics API integrations for tracking.
- Accounting software APIs for financial data export.

3.9 Deployment Architecture

Infrastructure:

- Frontend hosted on Vercel for global CDN distribution and automatic scaling.
- Backend services deployed on Vercel serverless functions.
- Database and authentication managed by Supabase cloud infrastructure.
- Static assets (images, CSS, and JS) are served via a CDN.

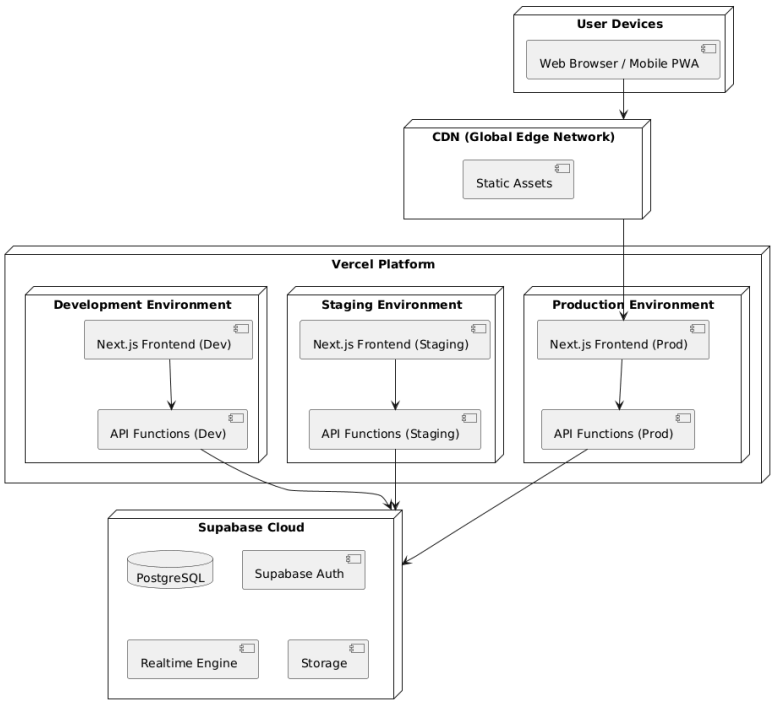
Environment Strategy:

- Development environment for feature development and testing.
- Staging environment mirroring production for final validation.
- Production environment with high availability and disaster recovery.

Scalability Approach:

- Horizontal scaling of serverless functions based on demand.
- Database connection pooling to handle concurrent requests.
- Caching layer (Redis) for frequently accessed data.
- CDN for static content delivery and reduced latency.

This architecture provides a robust, scalable foundation for IRIS, addressing the specific needs of fashion SMEs while maintaining flexibility for future enhancements and integrations. The full Deployment Architecture is depicted below:



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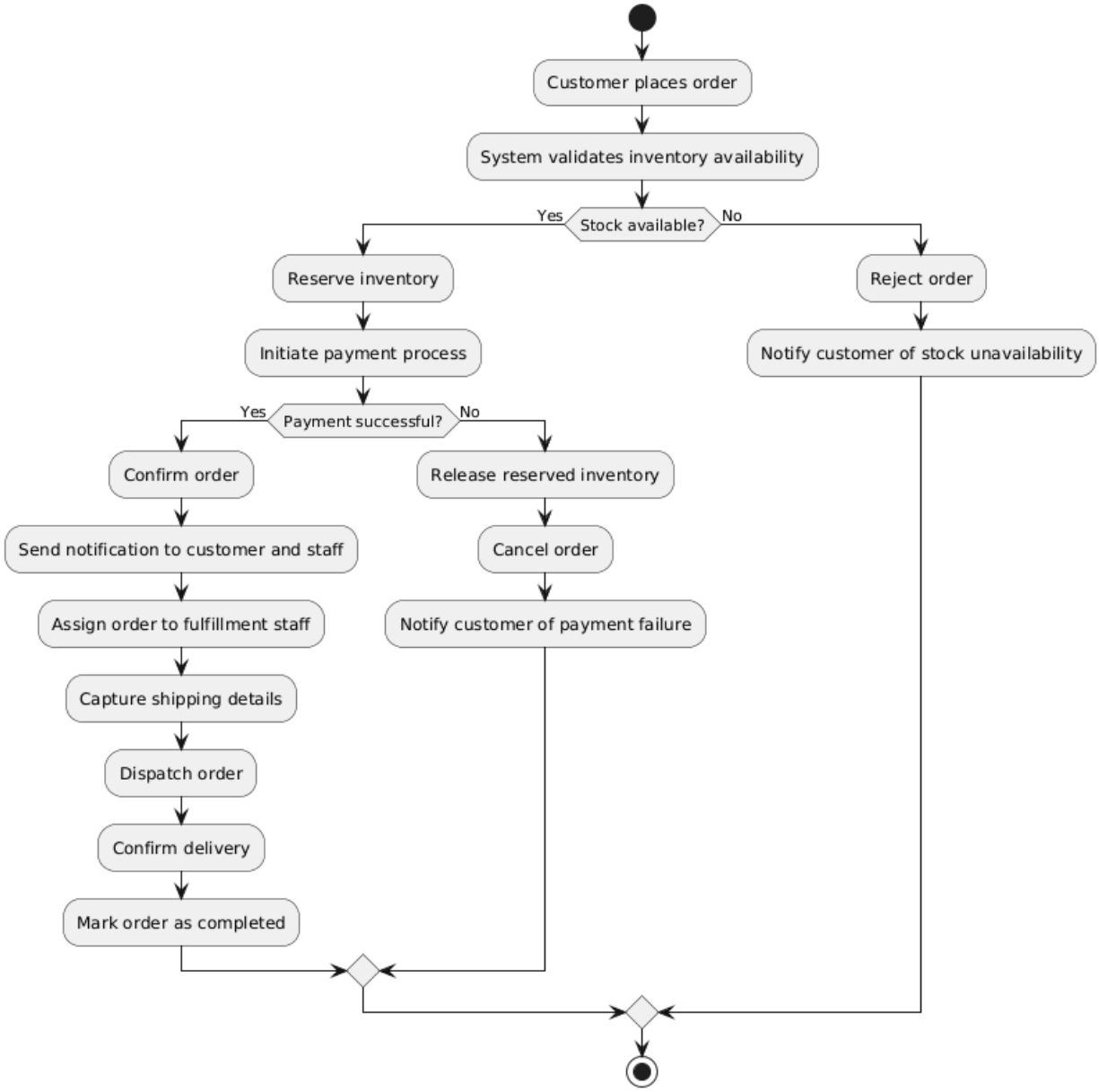
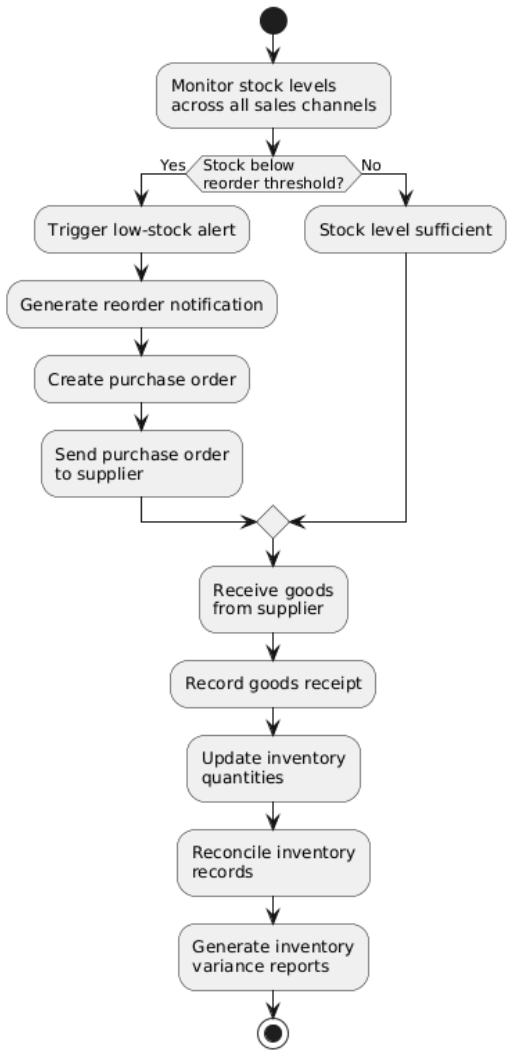
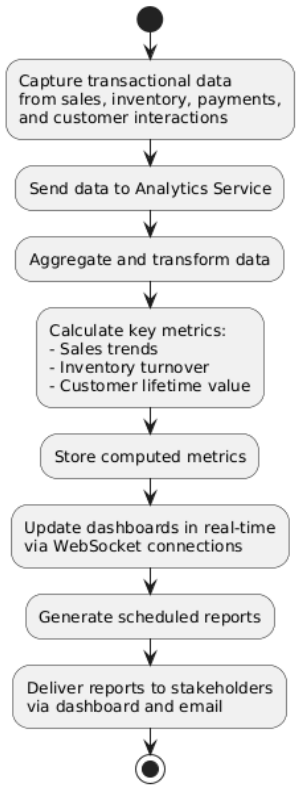
Appendices

Appendix 1: Order Fulfillment Workflow



Appendix 2: Inventory Management Workflow



Appendix 3: Analytics Generation Workflow